Extended Abstract

Motivation Current paradigms for humanoid whole-body control predominantly rely on policies trained to imitate dense, full-body reference trajectories. While effective in controlled environments, this approach suffers from two critical limitations. First, it imposes a **high specification burden**, as generating a complete, dynamically-feasible motion for the entire body is required for every new task. Second, such policies exhibit **poor adaptability**; by constraining every degree of freedom, the robot cannot easily deviate from the reference motion to react to unexpected perturbations or environmental variations. This motivates a shift towards a more practical control interface that is both sparse and adaptive. Our goal is to develop a foundational whole-body controller that operates using only sparse, task-critical commands—specifically the 6-DoF poses of the head and hands—liberating the robot to autonomously manage balance and generate the rest of its motion.

Method To address the challenges of sparse control, we introduce the **Sparse-Conditioned Whole-Body Controller (SC-WBC)**. The core of our method is a single-stage reinforcement learning (RL) framework that integrates a generative adversarial component to ensure the robot's autonomously generated motions are natural and human-like.

A key innovation is the use of **decoupled discriminators**. Instead of training a single discriminator on full-body poses, which would restrict the policy to reproducing motions seen in the reference dataset, we employ two separate discriminators: **A main-torso/lower-body discriminator**, trained to distinguish between the robot's locomotion style (torso, pelvis, legs) and clips of real human locomotion data.**An upper-body discriminator**, which performs the same function for the robot's manipulation movements (shoulders, elbows, wrists) relative to the torso. This decoupled architecture is critical for generalization. It prevents the entanglement of manipulation and locomotion skills, allowing the policy (the generator) to learn a versatile and compositional motion style. The policy is trained to fool both discriminators while simultaneously minimizing a goal-directed task reward that penalizes tracking error for the sparse head and hand targets.

Implementation The controller is trained exclusively in the **Isaac Gym** simulator and evaluated for sim-to-sim transfer in **MuJoCo**. We curated a **5-hour human motion dataset** (approx. 4,000 trajectories) from the AMASS and OMOMO datasets, which was retargeted to the G1 humanoid robot model. This provides a rich prior for both locomotion and manipulation. The final reward function is a composite of GAN-based style rewards and goal-directed tracking rewards. The additional auxiliary rewards to enforce physical plausibility, such as penalizing foot sliding and encouraging stable foot-ground contact has also been used.

Results We first highlight the discriminator decoupling design via quantitative experiments to evaluate our controller's performance on unseen motion tracking tasks against two baselines: a policy with a single full-body discriminator and one with no discriminator. Then, in the comparison of deepmimic and ICCGAN imitation strategies, we found the ICCGAN implicit style rewards has a higher potential in the sparse conditioned global tracking tasks.

Discussion Our qualitative analysis reveals why the decoupled discriminator architecture is so effective. A single, monolithic discriminator tends to entangle locomotion and manipulation patterns; for example, it may enforce an arm-swinging motion from walking data even when the task requires the robot to hold its arms steady to carry a box. Our decoupled design allows the policy to learn these skills compositionally, leading to more versatile and successful behaviors. Furthermore, we found that implicit, adversarial-style rewards are more suitable for global tracking tasks than explicit, joint-level imitation rewards. The latter can create conflicting objectives between matching a local reference pose and moving to a global target, whereas the adversarial reward implicitly guides the style without interfering with the primary task.

Conclusion We introduced the Sparse-Conditioned Whole-Body Controller (SC-WBC), a novel framework for intuitive and adaptive humanoid control. This work represents a significant step toward more practical, scalable, and adaptable humanoid controllers, lowering the barrier for task specification and enhancing the robot's ability to operate in dynamic, real-world environments.

Sparse-Conditioned Humanoid Whole-Body Controller

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Abstract

Current paradigms for humanoid whole-body control often rely on policies that imitate dense, full-body reference trajectories. This approach suffers from two critical limitations: a high specification burden, requiring complete, dynamically-feasible motions for every new task, and poor adaptability, as the policy is overly constrained and cannot easily react to perturbations. To address these issues, we introduce the Sparse-Conditioned Whole-Body Controller (SC-WBC), a novel framework that enables robust and natural humanoid control using only sparse end-effector commands. Our method conditions the policy on just three 6-DoF targets: the head pose for gaze and navigation, and the poses of both hands for manipulation. This drastically reduces the command complexity and liberates the robot's remaining degrees of freedom to autonomously discover movements for maintaining balance and achieving tasks. To ensure that the resulting motion is human-like, we integrate a generative adversarial component into a single-stage reinforcement learning pipeline. A key innovation is our use of decoupled discriminators, which separately assess the naturalness of upper-body and lower-body motions. This architecture prevents the entanglement of manipulation and locomotion skills, allowing the policy to learn a versatile and compositional motion style. Quantitative experiments demonstrate that our method achieves significantly lower tracking error and higher success rates on unseen motions compared to baselines using a single discriminator or no discriminator, while also showing superior sim-to-sim transferability. Our work provides a path toward more practical, adaptive, and scalable humanoid controllers.

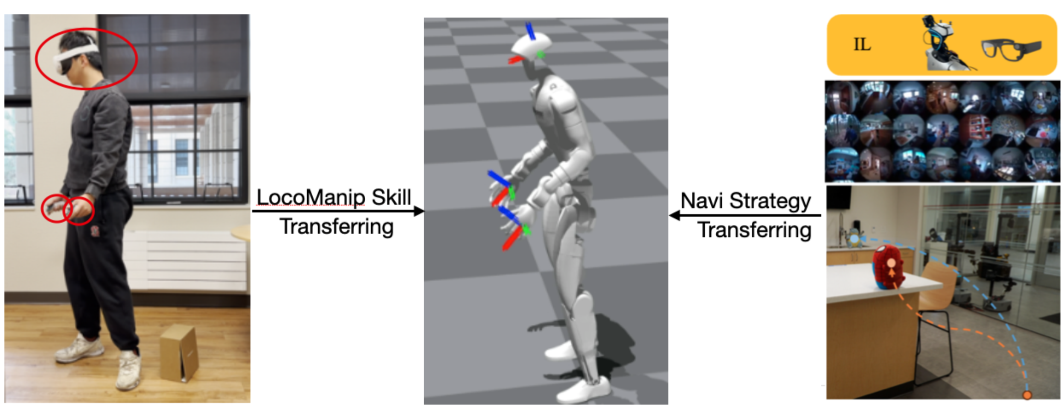

Figure 1: An illustration for 3-point loco-manipulation skill transferring and head-only navigation strategy transferring from human to humanoid, showing why the sparse 3-points with both position and orientation controls are the perfect interface for humanoid whole-body controller.

Stanford CS224R 2025 Final Report

1 Introduction

For many robotic platforms, such as fixed-base or wheeled manipulators, a robust control interface can be effectively realized through inverse kinematics (IK). An IK solver allows a user to command a task-space goal for an end-effector, providing a powerful and intuitive layer of abstraction. However, this paradigm is fundamentally insufficient for legged humanoid robots. Due to their floating base and high center of mass, humanoids are perpetually subject to gravity and must constantly manage ground contact forces to maintain balance. A purely kinematic solver is blind to these physical dynamics; it can generate joint configurations that, while kinematically valid, are dynamically unstable and would cause the robot to fall. Therefore, before any high-level learning or teleoperation can be achieved, a foundational **whole-body controller (WBC)** that explicitly considers the robot's full-body dynamics is an essential prerequisite.

The prevailing paradigm for developing such controllers relies on policies trained to imitate dense, full-body reference trajectories. These trajectories are typically sourced from human motion capture (mocap) or physics-based retargeting. While successful in controlled settings, this approach suffers from two critical limitations. First, it imposes a **high specification burden**, requiring a human operator or a separate planner to generate a complete, dynamically feasible motion for the entire body, a stark contrast to the simple end-effector commands common in manipulator robotics. Second, policies trained on such dense targets exhibit **poor adaptability**. Because the entire posture is constrained, the robot cannot deviate from the reference to accommodate unexpected perturbations or variations in the environment, such as changes in payload or terrain. An ideal controller should instead allow the robot to autonomously adjust its full-body posture to maintain balance and achieve the task, conditioned only on sparse, task-critical objectives.

To bridge this gap, we introduce a **Sparse-Conditioned Whole-Body Controller (SC-WBC)** for humanoid robots. Our method moves beyond dense joint-space targets, enabling control through only three sparsely specified end-effectors: the **head pose**, which directs the robot's gaze and navigational intent, and the **left and right hand poses**, which enable manipulation and interaction. By liberating the remaining DoFs, our controller allows the robot to autonomously discover and execute whole-body movements necessary to satisfy these sparse commands while dynamically maintaining balance.

A key challenge in this under-constrained setting is ensuring that the motion of the "free" joints is not just stable but also natural and human-like. Naive task optimization often produces jerky or idiosyncratic movements. We address this by incorporating a generative adversarial component into our training. A motion discriminator, trained to distinguish between real human motion clips and the policy's rollouts, provides a powerful adversarial reward. This encourages the policy to generate motions that are statistically similar to human examples, regularizing the behavior of the entire kinematic chain without requiring explicit, hand-engineered reward functions. Our approach is trained in a single stage, directly optimizing a policy that maps sparse commands to robust, natural, and task-aware whole-body motion.

The central contributions of this work are threefold:

- **A sparse-conditioned global-tracking whole-body controller** that tracks global end-effector targets for the head and hands. This significantly reduces the command burden and enables a more generalizable and adaptive control interface suitable for autonomous loco-manipulation tasks.
- **A single-stage training framework for loco-manipulation** that integrates adversarial imitation learning. This method efficiently produces a robust policy that simultaneously tracks sparse commands and exhibits natural, human-like motion, bypassing the need for complex multi-stage training or reward shaping.
- **An effective adaptation of animation techniques for robotic control**, demonstrating that generative models can replace complex, hand-tuned reward components common in robotics, leading to a simpler and more robust training pipeline.

2 Related Work

Whole-Body Control for Humanoid Robots A dynamic and robust whole-body controller (WBC) is the cornerstone of humanoid robotics, responsible for coordinating complex, high-DoF movements while maintaining balance. Classical approaches often rely on trajectory optimization and model-predictive control Mastalli et al. (2020); Kuindersma et al. (2016); Sreenath et al. (2011). While these methods can produce highly dynamic behaviors like jumping and traversing challenging terrain, they typically require a precise, pre-specified kinematic trajectory to track. Generating these trajectories is a significant challenge in itself, and they are often specific to a particular robot's morphology and the task at hand, limiting their generalizability.

To overcome the need for explicit trajectory generation, researchers have increasingly turned to reinforcement learning (RL), often in combination with imitation learning from human data. A prominent line of work trains policies to imitate dense, full-body motion capture data Peng et al. (2018); Luo et al. (2023); Fu et al. (2024); He et al. (2024b); Ji et al. (2024). These methods have demonstrated remarkable success in reproducing complex human skills on simulated and real robots. However, their reliance on full-body reference poses creates a significant bottleneck for practical applications, as it requires specialized equipment and comprehensive data for every new skill. This motivates a shift towards more accessible and less restrictive forms of human guidance.

Learning from Sparse Human Input Recent work aims to alleviate the burden of dense data capture by learning control policies from sparse inputs, which are easier to collect. For tabletop manipulation and navigation, sources like ego-centric video, hand poses, and gaze have proven effective Kareer et al. (2024); Wang et al. (2024); Chi et al. (2024); Cheng et al. (2024). This paradigm has been extended to humanoid teleoperation, where sparse signals from commercial VR/AR hardware serve as the control interface. For example, policies have been trained to track the position of the head and hands on VR headsets He et al. (2024a,c); Winkler et al. (2022), or to separately control the upper and lower body using VR and foot pedals Lu et al. (2024).

Our work builds on this foundation of using sparse head and hand tracking as the primary control interface He et al. (2024c,a), as it provides an intuitive bridge between human intent and robot action in the task space. However, we extend previous work in two crucial ways. First, we track not only the position but also the **full 6-DoF pose (position and orientation)** of the head and hands, enabling more dexterous and expressive loco-manipulation skills. Second, while most previous work focuses on local-frame teleoperation, our controller is designed for **global-frame task execution**, ensuring that the robot can accurately navigate to and interact with specific locations in its environment, a critical capability for autonomous task completion.

Insights from Physics-Based Character Animation The field of character animation has explored similar problems in controlling virtual characters for complex tasks like parkour or multi-agent games Tessler et al. (2024); Xu et al. (2023a,b); Xu and Wang (2024). Although these methods operate in clean, simulated environments without the complexities of robot hardware (e.g., sensor noise, domain gap, and retargeting errors), their underlying architectural choices offer valuable insights. For example, Tessler et al. (2024) introduced a unified control policy based on masked motion inpainting, demonstrating that plausible full-body motion can be reconstructed from only a partial reference signal. This concept of treating sparse control as a conditional motion generation problem directly inspires our approach.

However, a naive transfer of animation techniques to robotics is often unsuccessful. The absence of domain randomization and the challenges of human-to-humanoid retargeting can cause a significant performance drop. For instance, retargeting errors can introduce kinematic and dynamic inconsistencies that destabilize a policy trained only on clean mocap data Xu and Karamouzas (2021). To bridge this gap, our work explicitly incorporates robotics-centric rewards to enforce stable balance and proper foot-ground contact, grounding the generative power of animation-inspired techniques in the physical realities of humanoid control. This hybrid approach allows us to retain the architectural benefits of methods such as masked modeling while ensuring the robustness required for a real-world WBC.

3 Method

Our goal is to develop a whole-body controller that enables a humanoid to track sparse end-effector targets in the global frame while exhibiting natural, human-like motion. The control policy, π, takes as input the robot's current state and a set of target 6-DoF poses (position and orientation) for the head and hands. At each timestep, it outputs target joint positions that are executed by low-level PD servos. We formulate this as a reinforcement learning problem with a composite reward function designed to balance task success and motion style.

3.1 Adversarial Imitation with Decoupled Discriminators

To ensure the robot's movements are human-like, we employ an adversarial imitation learning framework inspired by ICCGAN Xu and Karamouzas (2021). Instead of training a single discriminator on full-body poses, which would restrict the policy to only reproducing motions seen in the reference dataset, we introduce a key architectural innovation: **decoupled discriminators**. As illustrated in Fig. 2, we use two separate discriminators:

- **A main-torso discriminator**, D_{lower}, which is trained to distinguish between the robot locomotion style (torso, pelvis, legs, feet) and clips of real human locomotion data for diverse balance strategies.
- **An upper-body discriminator**, D_{upper}, which performs the same function for the robot's upper body movements (shoulders, elbows, wrists – relatives to the torso). As any kinematically feasible upper body motion are also most likely be dynamically plausible, we train this discriminator with IK synthesized augmented dataset.

This decoupled design is critical for generalization. It allows the policy π (the generator) to synthesize new and creative behaviors by combining lower body movements from one reference motion with upper body movements from an entirely different one. The policy is rewarded for fooling both discriminators, encouraging it to learn a versatile motion style rather than simply memorizing fixed trajectories. The total style reward is a combination of the outputs of both discriminators.

3.2 Goal-Directed Control and Training Objective

While the adversarial component ensures natural motion, the policy must also accomplish its primary task of tracking sparse targets. We achieve this through a goal-directed reward term, r_{task}, which penalizes the error between the current and target 6-DoF poses of the head and hands in the global frame.

The final reward function combines style and task objectives. Following the multivalue design of ICCGAN Xu and Karamouzas (2021), we train a critic with separate heads to estimate the expected returns for the style and task rewards independently. This structure helps the actor (the policy π) to jointly optimize for both objectives during training, preventing one from dominating the other and leading to more stable learning. Additional rewards are included to enforce physically plausible behavior, such as maintaining balance and penalizing improper foot-ground contact.

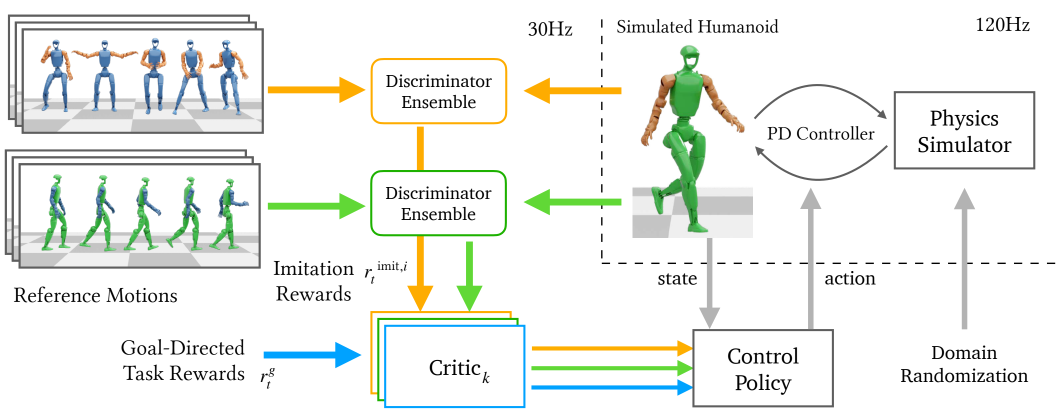

Figure 2: Systemic overview of the training scheme of our whole-body controller. We employ a multi-objective learning framework plus a GAN-based architecture for imitation learning. Our scheme allows imitating arm poses (orange) and those of the rest body parts (green) from different sources of reference motions simultaneously, and uses a goal-directed reward to fulfill the sparse tracking task for the head and hand poses.

3.3 Sim-to-real Considerations.

Our initial approach involved a single-stage, end-to-end policy learning pipeline adapted from ICCGAN. Within this framework, link-level information (e.g., positions and orientations) was essential for successful simulation-to-simulation transfer. However, this requirement introduced significant computational overhead, limiting the achievable control frequency on physical hardware.

To enable control at higher frequencies (i.e., >50 Hz), we transitioned to a codebase that had previously demonstrated sim-to-sim transferability for local keybody pose tracking using only joint-level observations. During the adaptation of this new framework, we conducted a qualitative analysis of several key factors, including the effects of hand-crafted foot-contact rewards, the comparison between DeepMimic-style and ICCGAN-style reward formulations, and the impact of the motion dataset size. This investigation helped establish a clearer roadmap toward our goal of developing a high-performance, sparse-condition, global-tracking humanoid whole-body controller.

4 Experimental Setup

Training and Evaluation Environments Our whole-body controller is trained exclusively in the Isaac Gym simulation environment. We then evaluate its performance in two distinct simulators: Isaac Gym Makoviychuk et al. (2021) itself and MuJoCo Todorov et al. (2012). Performance within Isaac Gym, as reported in Table 1 (left), directly reflects the impact of different design choices made during the training process. Conversely, the simulation-to-simulation evaluation in MuJoCo, shown in Table 1 (right), quantifies the policy's robustness and transferability, particularly as MuJoCo's contact dynamics are often considered a closer approximation of real-world physics Zhang et al. (2025). Unless otherwise specified, qualitative analyzes that compare different design choices are performed within the Isaac Gym environment.

Human Motion Dataset Curation To develop a versatile controller capable of a wide range of behaviors, we curated a 5-hour dataset (approximately 4,000 trajectories) by retargeting human motion capture data to the G1 robot. These data were sourced from the large-scale AMASS Mahmood et al. (2019) and OMOMO Li et al. (2023) datasets. The re-targetting process employs a keypoint matching technique similar to that described in He et al. (2024b). The resulting motion data set encompasses a rich set of representative behaviors that span both locomotion and manipulation tasks.

Furthermore, to isolate the effects of algorithmic design choices from the inherent difficulty of the tasks, our qualitative analysis also investigates the impact of these choices on a simplified problem: training a policy to track a single motion trajectory. This allows for a more controlled comparison of the tracking performance.

5 Results

5.1 Quantitative Evaluation

Design Choice	Isaac Gym - Unseen Motions			MuJoCo - Sim2Sim Transfer		
	Pos Error [m] $\downarrow$	Quat Error [rad] $\downarrow$	Failure [%] $\downarrow$	Pos Error [m] $\downarrow$	Quat Error [rad] $\downarrow$	Failure [%] $\downarrow$
Ours	**0.075**	**0.120**	**0**	**0.153**	**0.326**	**3**
Single discriminator	0.149	0.169	0	0.525	1.015	13
No discriminator	0.540	1.138	97	1.127	2.044	99

Table 1: Tracking accuracy is reported as positional error (m) and orientation error (rad). A timestep is counted as a failure when the head height deviates from the target by ≥ 0.4 m. Each configuration was trained for 50 k epochs, evaluated on 1-minute unseen motion clips, and repeated five times.

Single-stage RL Training Recipe. From Table. 1, we observed that guiding reinforcement learning (RL) exploration with generative adversarial networks (GANs) greatly improves sample efficiency. Compared to a single discriminator that jointly criticizes whole-body motion, we find that disentangling upper- and lower-body motion via separate reward from two discriminators helps. This is probably because separate discriminators prevent the policy from entangling irrelevant motions during training and achieve a lower tracking error. For instance, in most walking clips, people naturally swing their arms, while dual-arm manipulation typically occurs from a squatting posture. A policy trained with a *single* full-body discriminator tends to memorize the arm-swing pattern and then fails to handle a carried box while walking. In contrast, our setup decouples arm manipulation from balance control, enabling the whole-body controller to produce more diverse motions under 3-point tracking. In unseen tasks, our policy consistently outperforms the single whole-body discriminator variant.

5.2 Qualitative Analysis

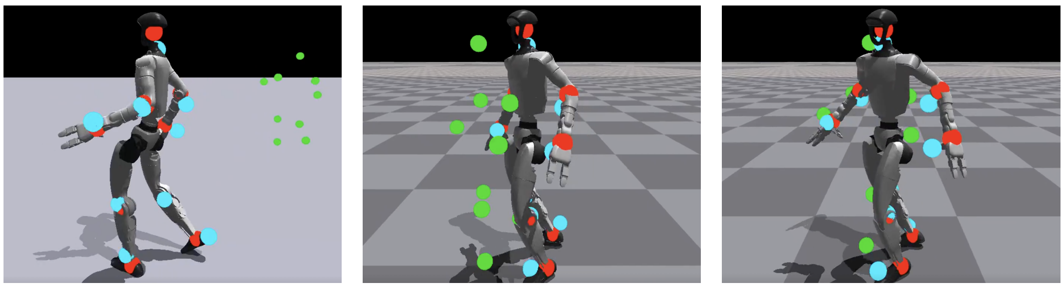

Figure 3: Left: Original agent performance for global tracking task from the teacher-student two-stage training pipeline with deepmimic-style rewards. Middle: Optimized agent that trained with more global tracking task rewards along with the deepmimic-style rewards. Right: Replace all joint-level tracking rewards with AMP(ICCGAN) discriminator score. Although including more global tracing rewards can help agent to track the green dots target in both(M, R) methods, the AMP ones (R) shows a higher potential for the accurate tracking.

The ICCGAN-style discriminator rewards have a higher potential for global tracking task than DeepMimic-style explicit joint level rewards Although deepmimic is more prevalent and easier to use for the full-body local tracking setup, it contains obvious limitations for the sparse-condition global tracking task. From Fig. 3, we observed that an obvious global tracking error persists for the agent trained with deepmic-style reward. The joint-level tracking rewards and the global keybody pose tracking rewards seem to be competing. Imagining a robot deviates from its global target pose but perfectly mimics the standing pose from the reference motion, it will still gain sufficient positive joint-level tracking rewards. Then, there is no incentive for the robot to break the standing pose and walk closer to the desired global goal pose, especially in the case where humanoid joint-level tracking is easier than the global keybody pose tracking task. Therefore, I believe that AMP-style rewards are essential for accurate global tracking tasks. AMP-style rewards implicitly how the learned policy acts similar to the reference motion, so it will not compete with the global pose tracking rewards as long as the generated motions are nature.

The additional foot-related rewards are helpful. For most of the re-targetting pipeline, we cannot fully eliminate the floating feet or ground penetration. When we leverage the GAN-based discriminator, the imperfect reference ankle motion will make the learned agent move in a wried patting manner. Compared to the efforts of manually fixing the reference motion, applying the additional foot-related rewards provides a shortcut for achieving robust ankle motions.

6 Discussion

Details to stabilize the GAN-based discriminator training. To leverage the power of ICCGAN, we must first enable stable GAN-based discriminator training. Unlike the MSE loss used in Peng et al. (2021), we found that the hinge loss mentioned in Xu and Karamouzas (2021) is more suitable for the case of using binary classification prob as implicit style rewards. Similarly, the details of computing the gradient penalty matter! Unlike some open-sourced navive AMP implementation on legged dog robots, we should use an interpolation of real and fake reference motion to euqally smooth the gradient onver the [-1, 1] score range. In Fig. 4, if we follow the naive implementation that only cares the smoothness mapping from all real reference data to lable of 1, the discriminator will converges to -1 and 1, lossing the purpose of providing gradual guidance to the RL explorations.

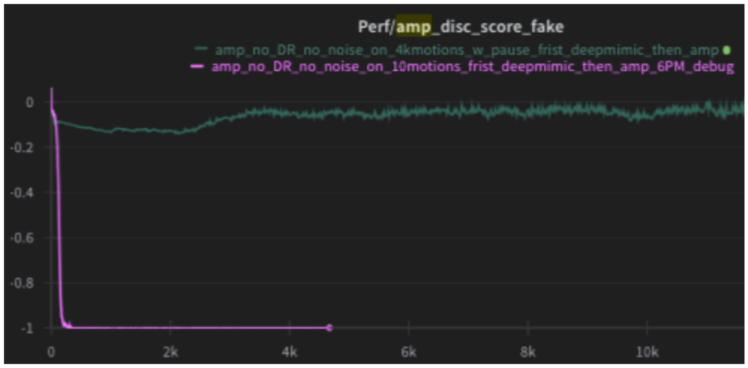

Figure 4: The comparison among two gradient penalty calculation over the prediction from discriminator about the fake dataset. If the gradient is too cliffy for any non-real reference dataset, it loss the purpose of providing gradual guidance to the RL explorations. In the ideal case, the discriminator should recognize every small improvements from full fake trajectories to nearly-real motions.

Combining Deepmimic and ICCGAN to learn faster Given the same amount of learning time, replacing the upper body discriminator by a set of explicit deepmimic joint-level tracking rewards can achieve better motion qualities. In Fig. 5, we can see that the arm represents an unnatural bending motion for the agent trained with implicit upper body regulation of the discriminator score. Considering that the upper body motion can be easily generated and achieve dynamically plausible quality, we do have access to an unlimited size of reference motion via the IK solver. In this case, using simple but effective deep-mimic joint-level tracking rewards can be an optimal option to achieve a faster learning process.

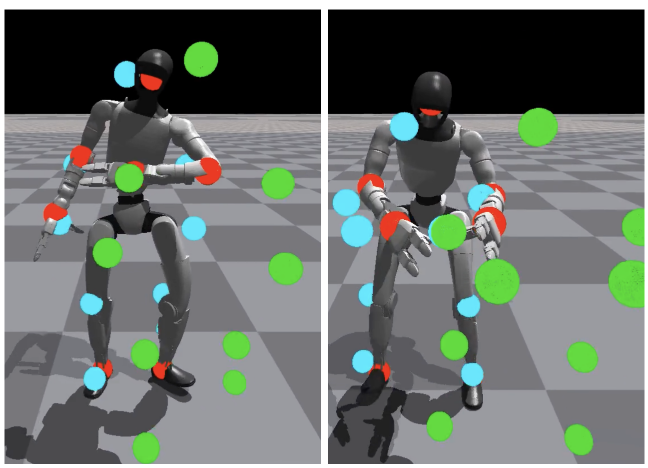

Figure 5: The comparison of upper body motion quality for agents trained with 5k epochs with (left) implicit upper body discriminator score and (right) explicit deepmimic joint-level tracking rewards.

Design choice of the termination penalty. To enforce the agent to prioritize the global tracking task, we intuitively terminate an episode if the robot is too far away from the global target. In naive implementation, we share the same termination penalty for failing down and the loss of track. However, from the ablation study shown in Fig. 6, we found that the policy can achieve better performance if we zero out the loss of track penalty but just reset the environment. In this manner, the robot will still try to eliminate the tracking error instead of giving up on this episode by falling directly. Meanwhile, as many times the drifting is not caused by the latest a few steps but leading by a far earlier decision, giving huge penalty to the latest actions does not make much sense. With frequent resets, the robot can have more chances of being exposed to the key-turning timing that causes the tracking error. Then, driven by the dominating global tracking rewards, the policy can achieve better performance in the zero loss of track penalty setting.

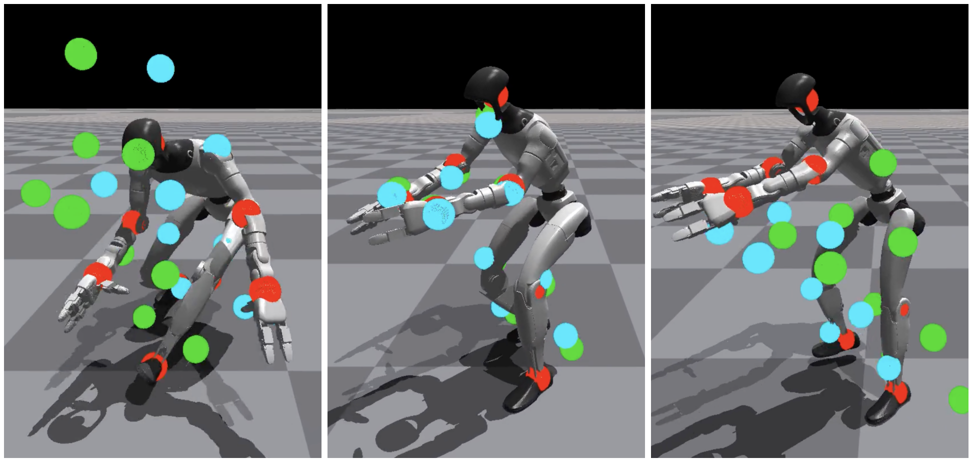

Figure 6: The effects of termination penalty when robot losses the track of global target has been shown in the (Left-A, Middle-B) sub figures. It's that the policy trained over 1 reference trajectory without loss of track penalty can achieve a better global tracking accuracy. By only changing the size of reference motion from 1 to 4000, we then get the agent shown on the most right-hand side (C). The comparison between B and C highlights how challenge the diverse global tracking task is. Also, it encourage us to identify each small design choice in a simply setting.

7 Conclusion

In this work, we presented the Sparse-Conditioned Whole-Body Controller (SC-WBC), a method designed to alleviate the high specification burden and poor adaptability of traditional humanoid controllers. By conditioning a policy on only sparse 6-DoF targets for the head and hands, our approach enables a more intuitive and flexible control interface suitable for complex loco-manipulation tasks. Our primary contribution includes a single-stage training framework that combines goal-directed rewards with an adversarial style reward derived from decoupled discriminators for the upper and lower body.

Our experiments demonstrated that this decoupled architecture is critical for success. It enables the policy to learn disentangled and compositional skills, leading to superior tracking performance and sim-to-sim robustness compared to alternative designs. We further showed through qualitative analysis that implicit, AMP-style adversarial rewards are more suitable for global tracking tasks than explicit, DeepMimic-style joint rewards, as they avoid conflicting optimization objectives. Our findings highlight the importance of architectural choices in the reward function and the training process to achieve sparsely conditioned high-performance control.

For future work, a promising direction is a two-stage training curriculum. First, a generalist policy could be pre-trained to master a wide variety of high-quality local motions using explicit, DeepMimic-style rewards. Subsequently, this expert local-tracking policy could be used to initialize or distill a second policy that is fine-tuned for the sparse, global-tracking task using our AMP-based framework. This approach could potentially stabilize GAN training by providing a strong motion prior, combining the motion quality of explicit imitation with the global task-awareness and flexibility of sparse adversarial imitation.

8 Team Contributions

- **Zi-ang Cao:** Create a humanoid RL codebase that can learn a sparsely conditioned whole-body controller for various humanoid motions. Study the performance gap in policy learning between using joint space observation and link space observation. Compare the global tracking potential of agents trained with deepmimic joint-level mimicing rewards and GAN implicit style rewards. Record the necessary tricks to make the policy more robust. For example, I find that the most effective way to make WBC be tolerant to delay is via post-training; if you consider the action delay from the first training epoch, the policy will converge to a bad motion quality. Explore the impacts of the data set on the final policy performance, emphasizing the details to learn GAN-based discremenators. From the ablation study, we also show a recommended way to design the discremenators.

Changes from Proposal We can achieve sparse conditioned WBC in two ways, through both single-stage training and teacher-student distillation. For the best motion quality, we conduct the comparison between deep-mimic explicity joint-mimicing rewards and GAN implicit style rewards on teacher policy.

References

An-Chieh Cheng, Yandong Ji, Zhaojing Yang, Zaitian Gongye, Xueyan Zou, Jan Kautz, Erdem Bıyık, Hongxu Yin, Sifei Liu, and Xiaolong Wang. 2024. Navila: Legged robot vision-language-action model for navigation. *arXiv preprint arXiv:2412.04453* (2024).

Cheng Chi, Zhenjia Xu, Chuer Pan, Eric Cousineau, Benjamin Burchfiel, Siyuan Feng, Russ Tedrake, and Shuran Song. 2024. Universal manipulation interface: In-the-wild robot teaching without in-the-wild robots. *arXiv preprint arXiv:2402.10329* (2024).

Zipeng Fu, Qingqing Zhao, Qi Wu, Gordon Wetzstein, and Chelsea Finn. 2024. Humanplus: Humanoid shadowing and imitation from humans. *arXiv preprint arXiv:2406.10454* (2024).

Tairan He, Zhengyi Luo, Xialin He, Wenli Xiao, Chong Zhang, Weinan Zhang, Kris Kitani, Changliu Liu, and Guanya Shi. 2024a. Omnih2o: Universal and dexterous human-to-humanoid whole-body teleoperation and learning. *arXiv preprint arXiv:2406.08858* (2024).

Tairan He, Zhengyi Luo, Wenli Xiao, Chong Zhang, Kris Kitani, Changliu Liu, and Guanya Shi. 2024b. Learning human-to-humanoid real-time whole-body teleoperation. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 8944–8951.

Tairan He, Wenli Xiao, Toru Lin, Zhengyi Luo, Zhenjia Xu, Zhenyu Jiang, Jan Kautz, Changliu Liu, Guanya Shi, Xiaolong Wang, et al. 2024c. Hover: Versatile neural whole-body controller for humanoid robots. *arXiv preprint arXiv:2410.21229* (2024).

Mazeyu Ji, Xuanbin Peng, Fangchen Liu, Jialong Li, Ge Yang, Xuxin Cheng, and Xiaolong Wang. 2024. Exbody2: Advanced expressive humanoid whole-body control. *arXiv preprint arXiv:2412.13196* (2024).

Simar Kareer, Dhruv Patel, Ryan Punamiya, Pranay Mathur, Shuo Cheng, Chen Wang, Judy Hoffman, and Danfei Xu. 2024. Egomimic: Scaling imitation learning via egocentric video. *arXiv preprint arXiv:2410.24221* (2024).

Scott Kuindersma, Robin Deits, Maurice Fallon, Andrés Valenzuela, Hongkai Dai, Frank Permenter, Twan Koolen, Pat Marion, and Russ Tedrake. 2016. Optimization-based locomotion planning, estimation, and control design for the atlas humanoid robot. *Autonomous robots* 40 (2016), 429–455.

Jiaman Li, Jiajun Wu, and C Karen Liu. 2023. Object motion guided human motion synthesis. *ACM Transactions on Graphics (TOG)* 42, 6 (2023), 1–11.

Chenhao Lu, Xuxin Cheng, Jialong Li, Shiqi Yang, Mazeyu Ji, Chengjing Yuan, Ge Yang, Sha Yi, and Xiaolong Wang. 2024. Mobile-television: Predictive motion priors for humanoid whole-body control. *arXiv preprint arXiv:2412.07773* (2024).

Zhengyi Luo, Jinkun Cao, Kris Kitani, Weipeng Xu, et al. 2023. Perpetual humanoid control for real-time simulated avatars. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 10895–10904.

Naureen Mahmood, Nima Ghorbani, Nikolaus F Troje, Gerard Pons-Moll, and Michael J Black. 2019. AMASS: Archive of motion capture as surface shapes. In *Proceedings of the IEEE/CVF international conference on computer vision*. 5442–5451.

Viktor Makoviychuk, Lukasz Wawrzyniak, Yunrong Guo, Michelle Lu, Kier Storey, Miles Macklin, David Hoeller, Nikita Rudin, Arthur Allshire, Ankur Handa, et al. 2021. Isaac gym: High performance gpu-based physics simulation for robot learning. *arXiv preprint arXiv:2108.10470* (2021).

Carlos Mastalli, Rohan Budhiraja, Wolfgang Merkt, Guilhem Saurel, Bilal Hammoud, Maximilien Naveau, Justin Carpentier, Ludovic Righetti, Sethu Vijayakumar, and Nicolas Mansard. 2020. Crocoddyl: An efficient and versatile framework for multi-contact optimal control. In *2020 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2536–2542.

Xue Bin Peng, Pieter Abbeel, Sergey Levine, and Michiel Van de Panne. 2018. Deepmimic: Example-guided deep reinforcement learning of physics-based character skills. *ACM Transactions On Graphics (TOG)* 37, 4 (2018), 1–14.

Xue Bin Peng, Ze Ma, Pieter Abbeel, Sergey Levine, and Angjoo Kanazawa. 2021. AMP: Adversarial Motion Priors for Stylized Physics-Based Character Control. *ACM Trans. Graph.* 40, 4, Article 1 (July 2021), 15 pages. `https://doi.org/10.1145/3450626.3459670`

Koushil Sreenath, Hae-Won Park, Ioannis Poulakakis, and Jessy W Grizzle. 2011. A compliant hybrid zero dynamics controller for stable, efficient and fast bipedal walking on MABEL. *The International Journal of Robotics Research* 30, 9 (2011), 1170–1193.

Chen Tessler, Yunrong Guo, Ofir Nabati, Gal Chechik, and Xue Bin Peng. 2024. MaskedMimic: Unified Physics-Based Character Control Through Masked Motion Inpainting. *ACM Transactions on Graphics (TOG)* (2024).

Emanuel Todorov, Tom Erez, and Yuval Tassa. 2012. Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on intelligent robots and systems*. IEEE, 5026–5033.

Chen Wang, Haochen Shi, Weizhuo Wang, Ruohan Zhang, Li Fei-Fei, and C Karen Liu. 2024. Dexcap: Scalable and portable mocap data collection system for dexterous manipulation. *arXiv preprint arXiv:2403.07788* (2024).

Alexander Winkler, Jungdam Won, and Yuting Ye. 2022. Questsim: Human motion tracking from sparse sensors with simulated avatars. In *SIGGRAPH Asia 2022 Conference Papers*. 1–8.

Pei Xu and Ioannis Karamouzas. 2021. A gan-like approach for physics-based imitation learning and interactive character control. *Proceedings of the ACM on Computer Graphics and Interactive Techniques* 4, 3 (2021), 1–22.

Pei Xu, Xiumin Shang, Victor Zordan, and Ioannis Karamouzas. 2023a. Composite Motion Learning with Task Control. *ACM Transactions on Graphics* 42, 4 (2023). `https://doi.org/10.1145/3592447`

Pei Xu and Ruocheng Wang. 2024. Synchronize Dual Hands for Physics-Based Dexterous Guitar Playing. In *SIGGRAPH Asia 2024 Conference Papers (SA '24)*. Association for Computing Machinery, New York, NY, USA, Article 143, 11 pages. `https://doi.org/10.1145/3680528.3687692`

Pei Xu, Kaixiang Xie, Sheldon Andrews, Paul G Kry, Michael Neff, Morgan McGuire, Ioannis Karamouzas, and Victor Zordan. 2023b. AdaptNet: Policy Adaptation for Physics-Based Character Control. *ACM Transactions on Graphics* 42, 6 (2023). `https://doi.org/10.1145/3618375`

John Z. Zhang, Taylor A. Howell, Zeji Yi, Chaoyi Pan, Guanya Shi, Guannan Qu, Tom Erez, Yuval Tassa, and Zachary Manchester. 2025. Whole-Body Model-Predictive Control of Legged Robots with MuJoCo. arXiv:2503.04613 [cs.RO] `https://arxiv.org/abs/2503.04613`

A Implementation Details for the single-stage RL recipe

A.1 Reward Terms

Style rewards. Instead of using a single discriminator for each motion group, we take the GAN-like architecture of ICCGAN Xu and Karamouzas (2021), and employ an ensemble of 32 discriminators for each motion group.

Task rewards. The goal-directed reward mainly measures tracking errors and also consists of three terms to stabilize the motions for sim-to-real consideration:

$$r_t^g = r_{\text{tracking}} + 0.8 r_{\text{in-air}} + 0.5 r_{\text{sliding}} + 0.005 r_{\text{energy}}. \tag{1}$$

For simplicity, here we omit the subscript t for each of the reward terms.

r_{tracking} is the reward measuring the tracking error:

$$r_{\text{tracking}} = 0.5 \exp\left(-\frac{5}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} |\varepsilon_{\text{pos},i}|\right) + 0.5 \exp\left(-\frac{2}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} |\varepsilon_{\text{orient},i}|\right) \tag{2}$$

where $\mathcal{I} = \{\text{left hand}, \text{right hand}, \text{head}\}$ is the set of links under tracking, $\varepsilon_{\text{pos},i}$ is the position error between the current position of the link i and its target position measured in Euclidean distance, and $\varepsilon_{\text{orient},i}$ is the orientation error measured by the angle between the orientation of the link i and the target.

$r_{\text{in-air}} = \min\{0, t_{\text{in-air}} - 0.5\}$ encourages the policy to keep the foot in the air for at least 0.5s during stepping. It is computed for the swing foot when it contacts the ground, and $t_{\text{in-air}}$ is the hanging time of that foot before the contact.

$r_{\text{sliding}} = -\sum_f c_f ||\mathbf{v}_f||_2$ penalizes the linear velocity of the foot link f if it contacts the ground, where $c_f = 1$ or 0 indicating the contact state of the foot f.

$r_{\text{energy}} = -\sum_j \left(0.1|\tau_j v_j| + 0.005\tau_j^2\right)$ penalizes the energy cost for each joint j, where τ_j is the torque applied on joint j and v_j is the joint's rotation velocity.

A.2 Domain Randomization

Parameters for domain randomization are listed in Table 2.

When testing the sim2real transfer, we found adding simulated delay during training time essential to prevent the robot from jittering. Compared to using a random delay across all environments, which is challenging to train, we found using a constant 1-step delay on a fixed portion ρ_{delay} of environments improves training speed, and can prevent the robot from jittering caused by the uncertainty delay during policy execution in the real world. We choose $\rho_{\text{delay}} = 0.5$, which means that the action delay is applied on half of the training environments.

Table 2: Domain Randomization Settings

Parameters	**Value**
Base Mass (kg)	$[-3, 3]$
Body Link Friction Coefficient	$[0.5, 1.25]$
Scale on PD Servo Gain (Kp)	$[0.7, 1.3]$
Scale on PD Servo Damping (Kd)	$[0.7, 1.3]$
Action Delay Ratio (ρ_{delay})	0.5